

Prompt 1 Initial Design Point Calculation

OTA 5T Initial Design Point Calculation (Prompt 1)

1. Role & Task Definition

You are an expert Analog IC Design Engineer AI Agent. Your task is to execute a step-by-step g_m/I_D methodology to systematically size a 5-Transistor Operational Transconductance Amplifier (5T OTA).

You will follow the mathematical guidelines and scripts provided below. You must not guess values or use trial-and-error; you must calculate the initial design point ($\vec{x}_{init}$) using the exact logic flow provided.

2. Important Note on Units (dB vs. Linear)

The target specifications for Voltage Gain (A_v) and Common-Mode Rejection Ratio (CMRR) provided to you in the spec sheet, as well as the outputs from the circuit simulator, are in **Decibels (dB)**.

However, the mathematical equations in this guideline use A_v and CMRR in **linear (normal) units**.

- **CRITICAL:** Before beginning any calculations, you must assume the input A_v and CMRR targets are in dB and explicitly convert them to linear units.

However, the mathematical equations in this guideline use A_v and CMRR in **linear (normal) units**.

Unit Conversion Formulas:

- **dB to Linear (Normal) Units:**

$$X_{\text{normal}} = 10^{\left(\frac{X_{\text{dB}}}{20}\right)}$$

- **Linear (Normal) Units to dB:**

$$X_{\text{dB}} = 20 \log_{10}(X_{\text{normal}})$$

- **CRITICAL INSTRUCTION:** Before beginning any g_m/I_D calculations, convert the input targets ($A_{v,\text{TG}}$ and CMRR_{TG}) from dB to linear units

using the conversion formula above. When reporting final simulation outputs, ensure you compare dB specs to dB simulator results directly.

3. Identify the Mathematical Problem

Circuit Feature Variables (Inputs):

$$\vec{x} = [W_d, L_d, W_m, L_m, W_{ss}, L_{ss}, W_B, I_B, C_L]$$

Circuit Target Variables (Outputs):

$$\vec{y} = f(\vec{x}) = [A_{vd}, BW, UGF, CMRR, PM, P_W, SR, A_{vd}(V_{iCM} = V_{iCM,TG,min}), A_{vd}(V_{iCM} = V_{iCM,TG,max})]$$

Variable Definitions:

- A_{vd} : Circuit Differential Gain
- BW: Circuit Bandwidth
- UGF: Circuit Unity Gain Frequency
- CMRR: Circuit Common Mode Rejection Ratio
- PM: Phase Margin of circuit
- P_W : Circuit power consumption
- SR: Circuit Slew Rate
- $A_{vd}(V_{iCM} = V_{iCM,TG,min})$: Circuit gain when we use $V_{iCM} = V_{iCM,TG,min}$
- $A_{vd}(V_{iCM} = V_{iCM,TG,max})$: Circuit gain when we use $V_{iCM} = V_{iCM,TG,max}$
- $V_{outminerr}$: When you put the circuit in buffer configuration & use $V_{in} = V_{out,TG,min}$, you get $V_{out} = V_{out,actual} \cdot V_{outminerr} = V_{out,actual} - V_{out,TG,min}$
- $V_{outmaxerr}$: When you put the circuit in buffer config & use $V_{in} = V_{out,TG,max}$, you get $V_{out} = V_{out,actual} \cdot V_{outmaxerr} = V_{out,actual} - V_{out,TG,max}$
- V_{outCM} : Output DC level when the circuit is biased at normal V_{iCM} & left in open loop config.

Specs Required to Achieve (Target Variable Specs):

$$\begin{aligned}
A_{vd} &\geq A_{v,TG} \\
UGF &\geq UGF_{TG} \\
CMRR &\geq CMRR_{TG} \\
PM &\geq PM_{TG} \\
P_W &\leq P_{W,TG} \\
V_{iCMmin} &\leq V_{iCM,TG,min} \implies A_{vd}(V_{iCM} = V_{iCM,TG,min}) \geq 0.8A_{v,TG} \\
V_{iCMmax} &\geq V_{iCM,TG,max} \implies A_{vd}(V_{iCM} = V_{iCM,TG,max}) \geq 0.8A_{v,TG}
\end{aligned}$$

4. Forced Conditions & Mission

You are forced to use a given Load Capacitance and Bias Current:

- $C_L = \text{given C value}$
- $I_B = \text{given Current value}$

Your feature variables reduce to:

$$\vec{x} = [W_d, L_d, W_m, L_m, W_{ss}, L_{ss}, W_B]$$

Mission: Find $\vec{x}_{opt}$ such that $f(\vec{x}_{opt}) \approx \vec{y}_{TG}$. You are required to use this step-by-step guide and tools to find $\vec{x}_{init}$.

5. Circuit Equations

1. $UGF = \frac{g_{m1}}{2\pi C_L}$
2. $A_{vCM} = \frac{1}{2g_{m3}r_{o5}}$
3. $A_{vd} = g_{m2}(r_{o2}||r_{o4}) \approx \frac{1}{2}g_{m1}r_{o1}$
4. $V_{iCMmax} = (V_{DD} - V_{SG3}) + V_{th1}$
5. $V_{iCMmin} = V_{GS1} + V_5^*$
6. $V_5^* = \frac{2}{(g_m/I_D)_5}$

Helper Tools

In the file called `LUT_FINDER_TOOL.py`, you will find all the finder functions (powered by the `LUT_Finder`) that you will need to use to query specific transistor parameters.

Use the python environment called `Thesis_Optimizer_Sandbox_Copy`

```

from LUT_FINDER_TOOL import LUT_Finder
finder = LUT_Finder("gf180_lut_data.npz")
from LTspice_Circuit_Wrappers import evaluate_5t_ota_ltspice
from SpiceFunctions import SpiceEvaluator , SpiceWaveformReader
LTSPICE_PATH =
r"C:\Users\Mohamed\AppData\Local\Programs\ADI\LTspice\LTspice.exe"
BASE_DIR =
r"C:\Users\Mohamed\Desktop\ML\CAD_projects\Netlists\GF_180nm_tech_work\project"

```

6. Step-by-Step Design Guide

Step 0: Overspecing

To get the design correctly you need to overspec at the start, to be in the safe zone from the beginning:

$$\begin{aligned}
A_{v,TG} &:= A_{v,TG} \times 1.1 \quad (+10\% \text{ over}) \\
UGF_{TG} &:= UGF_{TG} \times 1.1 \quad (+10\% \text{ over}) \\
CMRR_{TG} &:= CMRR_{TG} \times 1.1 \quad (+10\% \text{ over}) \\
V_{iCMTGmin} &:= V_{iCMTGmin} \times 0.95 \quad (-5\% \text{ over}) \\
V_{iCMTGmax} &:= V_{iCMTGmax} \times 1.05 \quad (+5\% \text{ over})
\end{aligned}$$

Keep P_W & PM as they are.

Step 1: Design M_1, M_2 (Differential Pair)

Start with UGF. We know that:

$$\frac{1}{C_L} \ll \frac{1}{C_{gg}} \implies \left(\frac{g_{m1}}{2\pi C_L} \right) \ll \left(\frac{g_{m1}}{2\pi C_{gg}} \right) \implies UGF \ll f_{T1}$$

Use: $f_{T,Q} = f_{T1} = 1000 \times UGF_{TG}$

Go to A_{vd} :

$$A_{vd} = \frac{1}{2}(g_m r_o)_1 \implies (g_m r_o)_1 = 2 \times A_{vd}$$

Use: $(g_m r_o)_1 = 2 \times A_{v,TG}$

Use the tools given to you (Don't leave M_1 yet, we continue to find some other stuff we will need later):

- `L1=finder.find_L_given_fT_gmgds(fT_Q=fT1, gm_gds_Q=gmro1, mos_type='nmos')`
- `ID_W1=finder.find_ID_W_given_L_gmgds(L_Q=L1, gm_gds_Q=gmro1, mos_type='nmos')`
- `gm_ID1=finder.find_gm_ID_given_L_gmgds(L_Q=L1, gm_gds_Q=gmro1, mos_type='nmos')`
- `VGS1 = finder.find_VGS_given_L_gmID( L_Q=L1, gm_ID_Q=gm_ID1, mos_type='nmos')`
- `Vth1 = finder.find_Vth_given_L_gmID( L_Q=L1, gm_ID_Q=gm_ID1, mos_type='nmos')`

Back to UGF:

$$g_{m1} = 2\pi \times C_L \times \text{UGF}_{TG}$$

$$I_{D1} = \frac{g_{m1}}{(g_m/I_D)_1}$$

$$W_1 = \frac{I_{D1}}{(I_D/W)_1}$$

Set Features:

- $W_d = W_1$
 - $L_d = L_1$
- (Obtained 2 out of 7 features)*

Step 2: Design M_3, M_4 (Mirror Devices)

From the diff pair: $I_{D3} = I_{D1}$ and $r_{o3} = r_{o1}$.

(Reminder: we already have I_{D1} and g_{m1} calculated before).

$$A_{v,TG} = \frac{1}{2}(g_m r_o)_1 = \frac{1}{2}g_{m1}r_{o1} \implies r_{o1} = \frac{2A_{v,TG}}{g_{m1}}$$

We have obtained r_{o1} , so $r_{o3} = r_{o1}$. We can obtain:

$$\left(\frac{g_{ds}}{I_D}\right)_3 = \frac{1}{r_{o3}I_{D3}} = \frac{1}{r_{o1}I_{D1}}$$

Go to $V_{iCMTGmax}$:

$$V_{iCMTGmax} = (V_{DD} - V_{SG3}) + V_{th1} \implies V_{SG3} = V_{DD} - V_{iCMTGmax} + V_{th1}$$

Now go to your tools:

- `L3=finder.find_L_given_gdsID_VGS(gds_ID_Q=gds_ID3, VGS_Q=VSG3, mos_type='pmos')`

- `gm_ID3=finder.find_gm_ID_given_L_VGS(L_Q=L3, VGS_Q=VSG3, mos_type='pmos')`

- `ID_W3=finder.find_ID_W_given_L_gmID(L_Q=L3, gm_ID_Q=gm_ID3, mos_type='pmos')`

Calculate width:

$$W_3 = \frac{I_{D3}}{(I_D/W)_3}$$

Calculate g_{m3} (will be used later):

$$g_{m3} = (g_m/I_D)_3 \times I_{D3}$$

Set Features:

- $L_m = L_3$
- $W_m = W_3$
(Obtained 4 out of 7 features)

Step 3: Design M_5, M_6 (Tail Device & Mirror)

From $V_{iCMTGmin}$:

$$V_{iCMTGmin} = V_{GS1} + V_5^* \implies V_5^* = V_{iCMTGmin} - V_{GS1}$$

$$(g_m/I_D)_5 = \frac{2}{V_5^*}$$

$$I_{D5} = I_{ss} = 2 \times I_{D1}$$

From CMRR:

$$A_{vCM} = \frac{A_{v,TG}}{CMRR_{TG}}$$

$$A_{vCM} = \frac{1}{2g_{m3}r_{o5}} \implies r_{o5} = \frac{CMRR_{TG}}{2g_{m3}A_{v,TG}}$$

We have I_{D5} and r_{o5} , so:

$$\left(\frac{g_{ds}}{I_D} \right)_5 = \frac{1}{r_{o5}I_{D5}}$$

Use tools:

- ```
L5=finder.find_L_given_gdsID_gmID(gds_ID_Q=gds_ID5,
gm_ID_Q=gm_ID5, mos_type='nmos')
```
- ```
ID_W5=finder.find_ID_W_given_L_gmID( L_Q=L5, gm_ID_Q=gm_ID5,  
mos_type='nmos')
```

Calculate widths:

$$W_5 = \frac{I_{D5}}{(I_D/W)_5}$$

Set Features:

- $W_{ss} = W_5$
- $L_{ss} = L_5$

Since M_6 is a mirror of M_5 :

$$\frac{W_{ss}}{W_6} = \frac{I_{ss}}{I_B} = \frac{I_{D5}}{I_B} \implies W_6 = \frac{W_6 \cdot I_B}{I_{ss}} = \frac{W_5 \cdot I_B}{I_{D5}}$$

Set Feature:

- $W_B = W_6$

(Obtained 7 out of 7 features)

7. Restrictions & Final Execution

1. You aren't allowed to change any of the tools or files that are already in the project folder except for the file called `workhere.py` or the notebook called `workin_here.ipynb`. These are empty files made specifically for you to use and write your code, math, notes, and implementation. You can create your own python files, Jupyter notebooks, notepad, text files,...etc if needed.
2. Follow the step-by-step design guide to achieve the specs and try not to improvise.
3. Any mathematical calculations or numbers should be done on Python and using Python tools only. Don't try to calculate on your own (you can use NumPy if needed).
4. Write your outputs `[W_d, L_d, W_m, L_m, W_ss, L_ss, W_B]` in a `.py` file like this:

```
W_d = ...
L_d = ...
W_m = ...
L_m = ...
W_ss = ...
L_ss = ...
W_B = ...
```

5. After you are done with the initial design point calculations you must run a single simulation run using the OTA circuit wrapper function `evaluate_5t_ota_ltspace(Wd, Ld, Wm, Lm, Wss, Lss, WB, Ib, CL, Vicm, TGT_VICM_MIN, TGT_VICM_MAX, TGT_VOUT_MIN, TGT_VOUT_MAX, BASE_DIR='.')` that is given to you inside the python file called `LTspice_Circuit_Wrappers.py`.


```

LTSPICE_PATH =
r"C:\Users\Mohamed\AppData\Local\Programs\ADI\LTspice\LTspice.exe"
BASE_DIR =
r"C:\Users\Mohamed\Desktop\ML\CAD_projects\Netlists\GF_180nm_tech_work\project"

TGT_VICM_MIN=Vicmmin # VicmTGmin given in the spec
TGT_VICM_MAX=Vicmmax # VicmTGmax given in the spec

TGT_VOUT_MIN=0.5 # Just use this number since no spec on it (Any number
to run the Wrapper)
TGT_VOUT_MAX=2.5 # Just use this number since no spec on it (Any number to
run the Wrapper)

# The design parametres you just obtained
Ld= L1
Wd= W1
Lm= L3
Wm= W3
Lss= L5
Wss= W5
WB = W6

Ib=given_Current_number # use the value that will given in the spec file
CL=given_number # use the value that will given in the spec file
Vicm= Vicm_normal # given in the spec file

specs=evaluate_5t_ota_ltpice(Wd, Ld, Wm, Lm, Wss, Lss, WB, Ib, CL, Vicm,
                             TGT_VICM_MIN, TGT_VICM_MAX, TGT_VOUT_MIN,
                             TGT_VOUT_MAX,
                             BASE_DIR=BASE_DIR)

print("\n" + "=" * 65)

```

```

print(" 5T OTA SPECIFICATION GENERATOR")
print("=" * 65)
print(f" 1.  Av                : {specs['Av']:.4f} dB")
print(f" 2.  BW                 : {specs['BW']/1e3:.2f} kHz")
print(f" 3.  UGF                 : {specs['UGF']/1e6:.2f} MHz")
print(f" 4.  CMRR               : {specs['CMRR']:.4f} dB")
print(f" 5.  PM                 : {specs['PM']:.2f} deg")
print(f" 6.  Power              : {specs['Pw']*1e6:.2f} uW")
print(f" 7.  Slew Rate          : {specs['Slew_Rate']/1e6:.2f} V/us")
print(f" 8.  Av(@TGT_VICM_MIN) : {specs['Av_VICM_MIN']:.2f} dB")
print(f" 9.  Av(@TGT_VICM_MAX) : {specs['Av_VICM_MAX']:.2f} dB")
print(f"10.  VOUT_MIN Err       : {specs['VOUT_MIN_Err']*1000:.2f} mV")
print(f"11.  VOUT_MAX Err       : {specs['VOUT_MAX_Err']*1000:.2f} mV")
print(f"12.  VOUT_CM            : {specs['VOUT_CM']:.2f} V")
print("=" * 65)

```

6. Report the output results in the chat and in a notepad or text file.
7. Mention whether all the specs were achieved or if some specs haven't been achieved, mention this in both the chat and the notepad/txt file , in case some of the specs were not achieved , mention them in both the chat and the notepad/txt file .

The results in both the chat and the notepad should be in a table like this

Spec	Target	Achieved	Status
Av			[PASS/FAIL]
UGF			[PASS/FAIL]
CMRR			[PASS/FAIL]
PM			[PASS/FAIL]
Pw			[PASS/FAIL]
Av_VICM_MIN			[PASS/FAIL]
Av_VICM_MAX			[PASS/FAIL]

=====